

KFUPM Maps

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a. Project Overview

Project Title: KFUPM Maps – Indoor & Outdoor Navigation Web Application

Executive Summary:

KFUPM Maps is an interactive web-based navigation system designed to help users find routes between buildings across the KFUPM campus. The system provides step-by-step photo-based directions, a searchable campus map, a leaderboard for active users, and administrative tools for managing routes. The application consists of a React/Vite frontend and a Node.js/Express backend with MongoDB as the primary database.

b. Technologies and Tools

- **React.js (with Vite):** Used to build the single-page frontend application and manage UI components.
- **React Router:** Handles client-side routing between pages such as Home, View Route, Leaderboard, and Account.
- **CSS / Custom Styling:** Provides layout, colors, spacing, and responsive design for different screen sizes.
- **Node.js:** JavaScript runtime used to execute backend logic on the server.
- **Express.js:** Web framework used to define RESTful API endpoints for authentication, routes, likes, and leaderboard.
- **MongoDB with Mongoose:** Stores users, routes, route steps, likes, and statistics in a NoSQL database.
- **JWT (JSON Web Tokens):** Implements secure user authentication and session handling across client and server.
- **bcrypt:** Hashes user passwords before storing them in the database to enhance security.
- **Vercel:** Deployment platform used to host the frontend React application.
- **Git & GitHub:** Used for version control, collaboration, and source code hosting.

c. GUI Screenshots

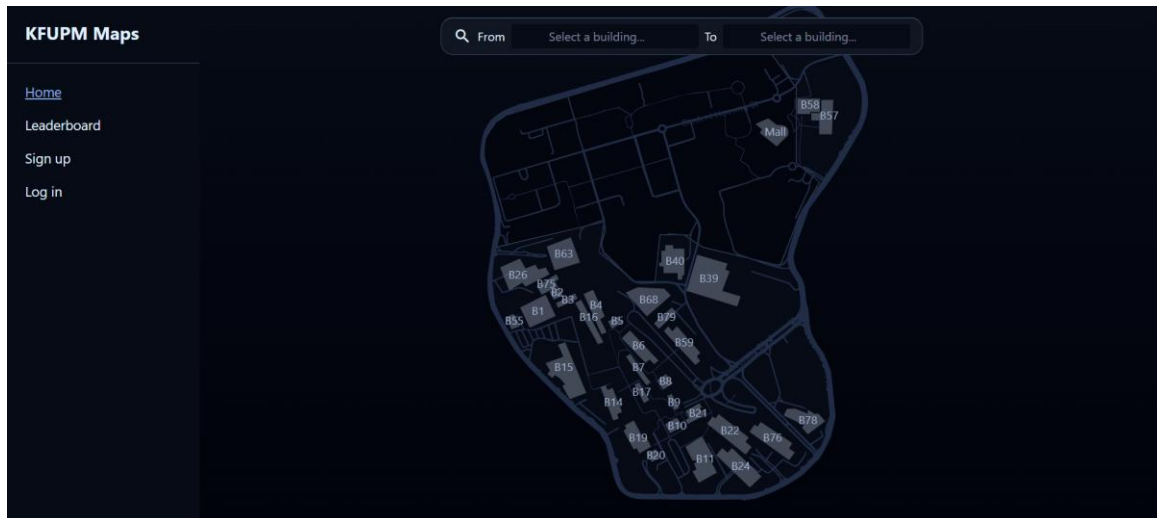


Figure 1: Home Page (campus map and building search).

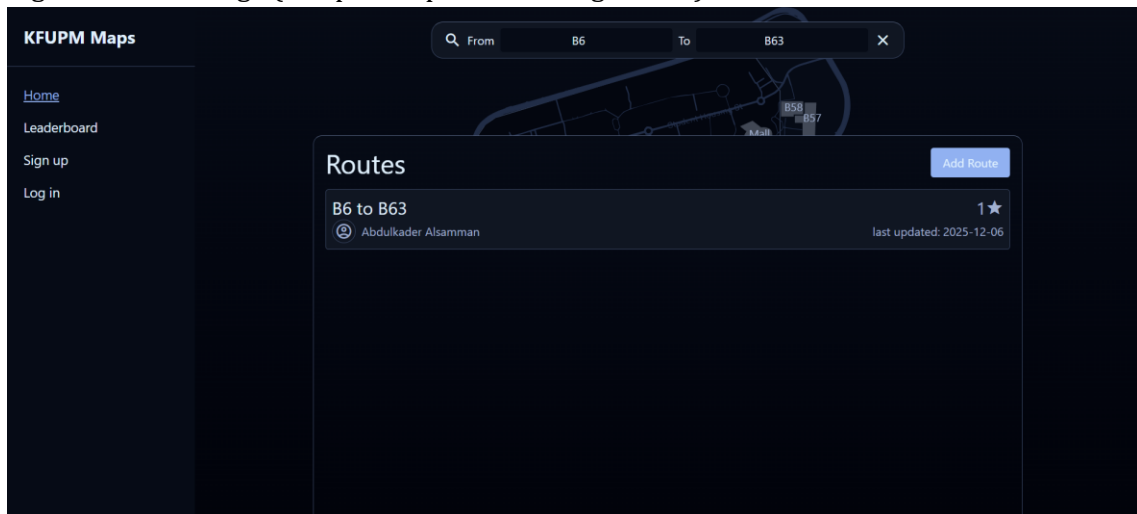


Figure 2: Home Page after selecting buildings and showing suggested routes.

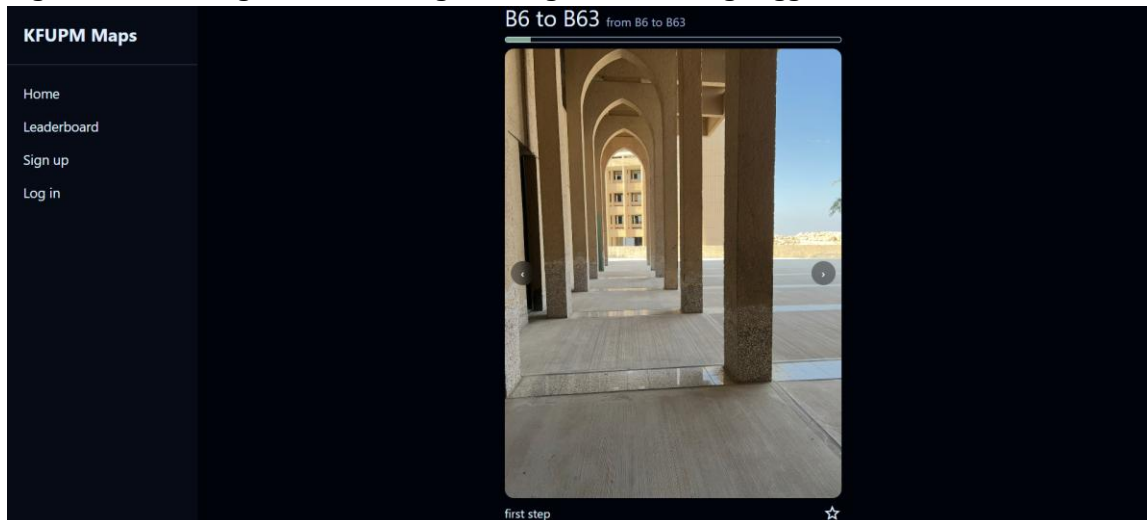


Figure 3: View Route Page with step-by-step image navigation.

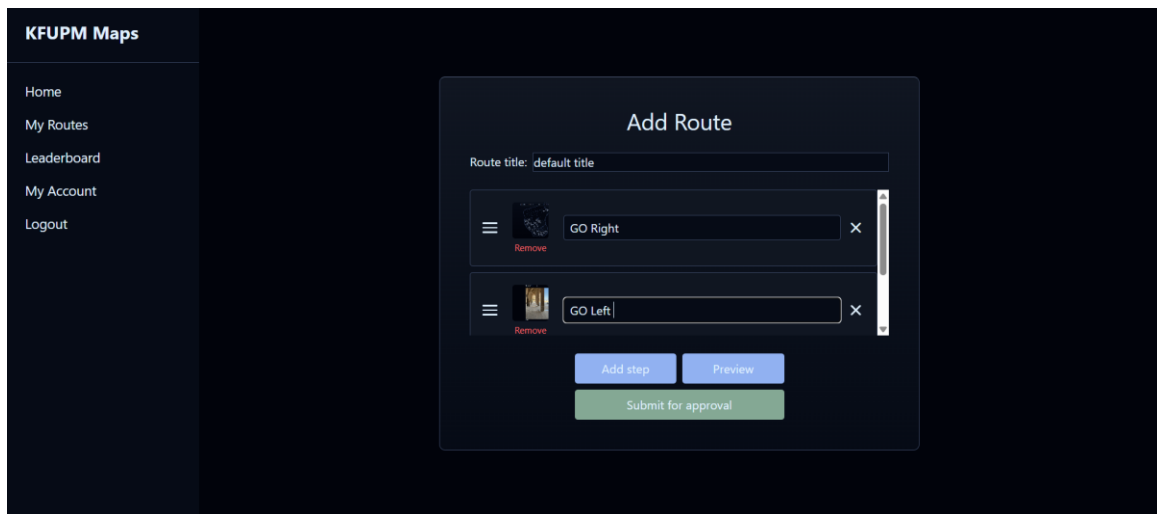


Figure 4: Add Route Page

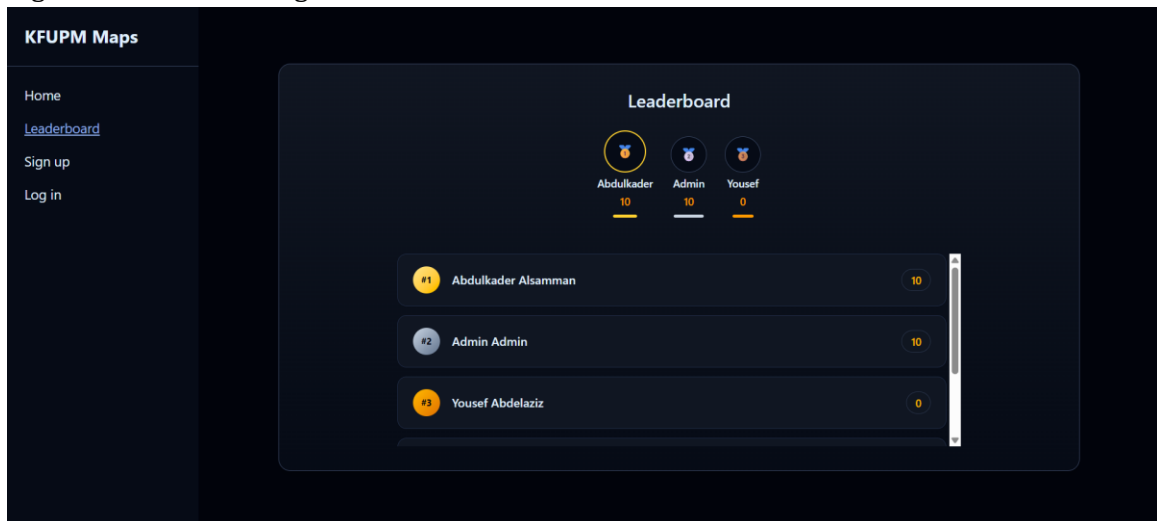


Figure 5: Leaderboard Page showing user rankings.

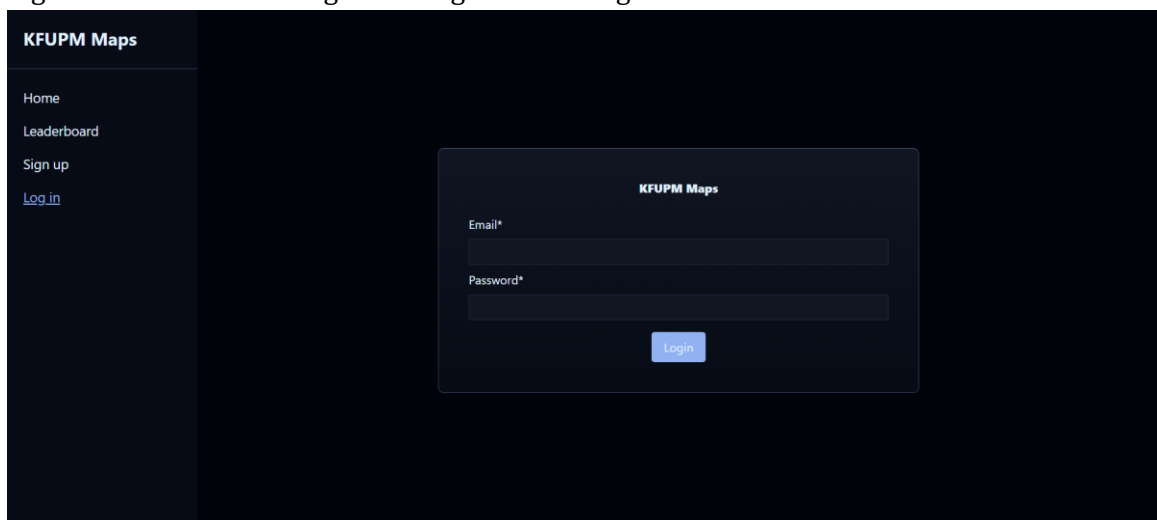


Figure 6: Login Page.

KFUPM Maps

Home

Leaderboard

[Sign up](#)

Log in

KFUPM Maps

First Name*

Last Name*

Email*

Password*

Sign up

Figure 7: Signup Page.

KFUPM Maps

Home

My Routes

Leaderboard

[My Account](#)

Logout

My Account

Update Photo

First Name

Mohammed

Last Name

Yar

Save

Figure 8: My Account / Profile Page.

KFUPM Maps

Home

My Routes

[Manage Routes](#)

Leaderboard

My Account

Logout

Pending

Approved

Rejected

default title	0 ★
Abdulkader Alsamman	last updated: 2025-12-06
default title	0 ★
Abdulkader Alsamman	last updated: 2025-12-06
default title	0 ★
Mohammed Yar	last updated: 2025-12-09
default title	0 ★
Mohammed Yar	last updated: 2025-12-09
default title	0 ★
Mohammed Yar	last updated: 2025-12-09

Figure 9: Manage Routes (Admin) Page.

d. Source Code Highlights

This section presents selected source code snippets that demonstrate core functionalities of the system.

Backend Authentication Token Generation

```
// utils/token.js
const jwt = require("jsonwebtoken");

const generateToken = (userId) => {
  return jwt.sign({ id: userId }, process.env.JWT_SECRET, {
    expiresIn: "7d",
  });
};

module.exports = generateToken;
```

This function generates a signed JSON Web Token (JWT) that is used to authenticate users after successful login. The token includes the user ID and an expiration time, allowing secure, stateless authentication.

Route Creation Logic (Backend)

```
// controllers/routeController.js
const createRoute = async (req, res) => {
  const { title, from, to, steps } = req.body;

  const route = new Route({
    title,
    from,
    to,
    steps: steps.map((step) => ({
      caption: step.caption,
      imageUrl: step.imageUrl,
    })),
  });

  await route.save();
  res.status(201).json(route);
};
```

This controller function creates a new route document in MongoDB. It receives a list of steps from the client, maps each step into an object containing a caption and image URL, and then saves the route to the database.

Route Step Viewer (Frontend)

```
// ViewRoute.jsx (excerpt)
const [currentIndex, setCurrentIndex] = useState(0);

const currentStep = route.steps[currentIndex];

return (
  <div className="route-step-container">
    <img src={currentStep.imageUrl} alt="Route step" />
    <p>{currentStep.caption}</p>
    <div className="step-controls">
      <button onClick={handlePrevious} disabled={currentIndex === 0}>
        Previous
      </button>
    </div>
  </div>
);
```

```

    <span>{currentIndex + 1} / {route.steps.length}</span>
    <button
      onClick={handleNext}
      disabled={currentIndex === route.steps.length - 1}
    >
      Next
    </button>
  </div>
</div>
);

```

This React component excerpt manages the current step index and displays the corresponding image and caption. Navigation buttons allow users to move forward and backward through the route steps.

Leaderboard Retrieval (Backend)

```

// controllers/leaderboardController.js
const getLeaderboard = async (req, res) => {
  const users = await User.find()
    .sort({ score: -1 })
    .limit(20)
    .select("name score");

  res.json(users);
};

```

This function retrieves the top users sorted by score from the database. The data is used on the frontend to display the leaderboard page.

e. Open Source Code Reusability

The project builds upon several open-source frameworks and libraries. All dependencies are included through the package.json files and retain their original licenses. The primary open-source components are listed below:

- **React.js:** MIT License – used as the main frontend framework.
- **Vite:** MIT License – used as the build and development tool for the React project.
- **Node.js / Express.js:** MIT License – used to implement the backend REST API.
- **MongoDB / Mongoose:** MongoDB Server Side Public License (SSPL) – used for data storage and object modeling.
- **jsonwebtoken:** MIT License – used for creating and verifying JWT tokens for authentication.
- **bcrypt:** MIT License – used to hash user passwords securely.

No third-party source files were directly copied into the project without attribution. All reused components are standard open-source libraries that are properly acknowledged through their inclusion as dependencies.

f. Learnings and Reflections

Key Learnings:

- Designing and implementing a full-stack web application using React on the frontend and Node.js/Express on the backend.
- Working with MongoDB and Mongoose to define schemas, relationships, and efficient queries.
- Implementing secure authentication flows using JWT and protecting routes on both client and server sides.
- Structuring a scalable API and organizing the backend into controllers, models, and utility modules.
- Deploying a production-ready application and understanding environment configuration for frontend and backend services.

Limitations:

- Routes are currently created and maintained manually by administrators; there is no automatic path calculation.
- The navigation experience depends heavily on the clarity and coverage of the route images provided.
- Offline usage is not supported; the application requires an active internet connection.
- Mobile responsiveness works but can be further refined for smaller screens and different orientations.

Future Improvements:

- Introduce automated pathfinding algorithms to generate optimal routes between buildings based on a campus graph.
- Support multi-floor indoor navigation with floor selection and more detailed maps.
- Enhance the UI/UX with additional accessibility features and theme options.
- Implement analytics to track popular routes and identify areas where more guidance is needed.

g. Team Contributions (Time-sheet)

Task	Member(s)	Period	Description
Frontend UI Design	All	1 Week	Designed the main layout, map interface, and page components.
Backend API Development	All	2 weeks	Implemented authentication, route management, likes, and leaderboard APIs.

Database Design & Integration	Yousef Abdulaziz & Mohammed Yar	3 days	Defined MongoDB schemas and configured Mongoose models.
Testing & Quality Assurance	Khaled Aljohani & Abdulkader Alsamman	2 days	Performed manual testing, identified bugs, and verified fixes.
Deployment & Environment Setup	Yousef Abdulaziz	4 hours	Configured and deployed the frontend and backend services.
Documentation & Final Report	All	6 hours	Compiled documentation, prepared the final report, and organized project artifacts.